

第一问 · 论文图册

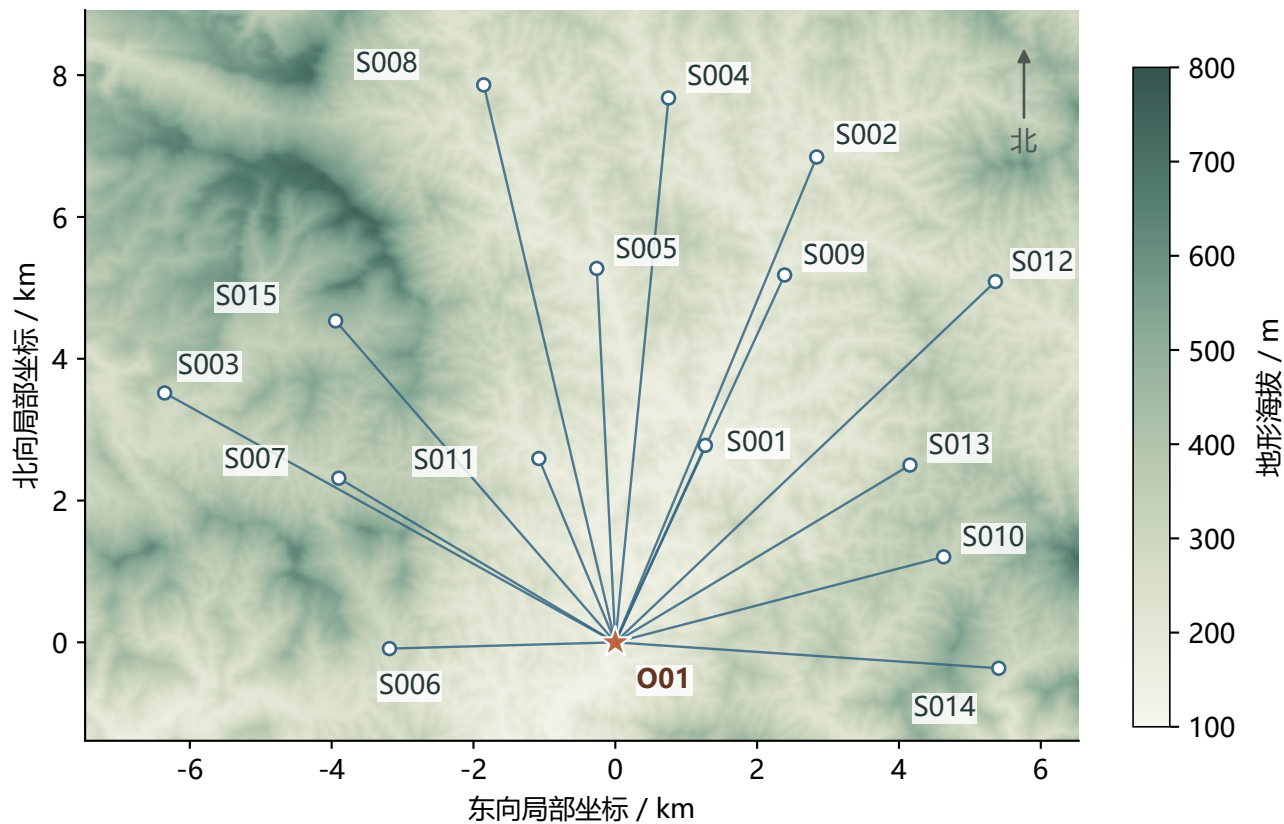
局部椭圆坐标 · 整箱组批 · 多目标权衡 · 精确余量阈值

16幅图 / 独立矢量PDF与SVG / 600dpi PNG / 配套源数据

F01	地形、节点及直达航线	正文优先
F02	关键航线的地形剖面	
F03	各服务区的物资需求	
F04	基准18架次的整箱组批方案	正文优先
F05	三类容量约束的利用率	
F06	往返运输能耗的分项组成	
F07	累计作业时间的分项组成	
F08	20%返航余量下的最大安全载荷	正文优先
F09	完整离散Pareto前沿	正文优先
F10	两个前沿方案的S008组批差异	
F11	安全余量变化下的精确最优方案阶梯	正文优先
F12	19段最优方案及指标对照	
F13	15服务区三类机型的安全载荷曲线	
F14	最终狭窄可行区间与机型切换	正文优先
F15	满载与空载的安全余量边界	
F16	第一问建模与精确求解流程	

基准：18架次 / 59.132260 kWh / 累计546.275280 min

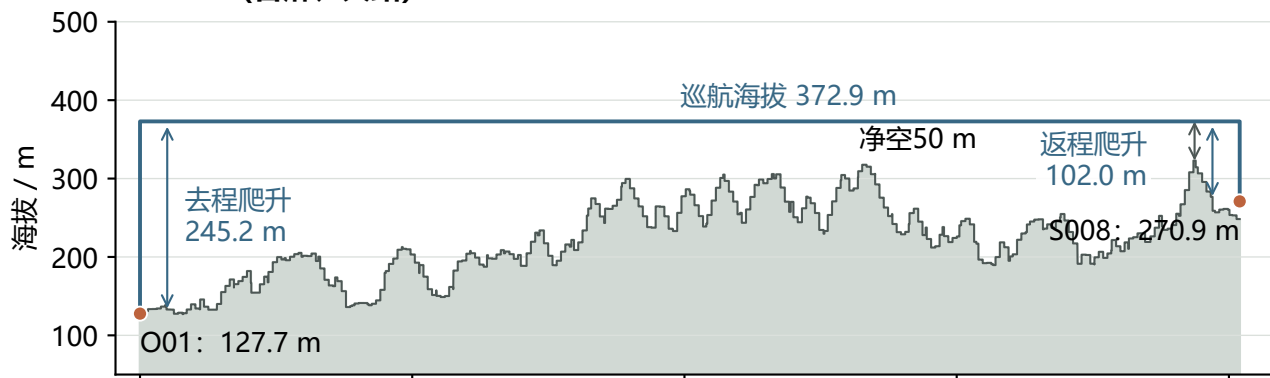
完整图注与解释见“文字/图表说明与选用建议.md”。
图中数值为给定模型下的计算结果；不代表实测精度。



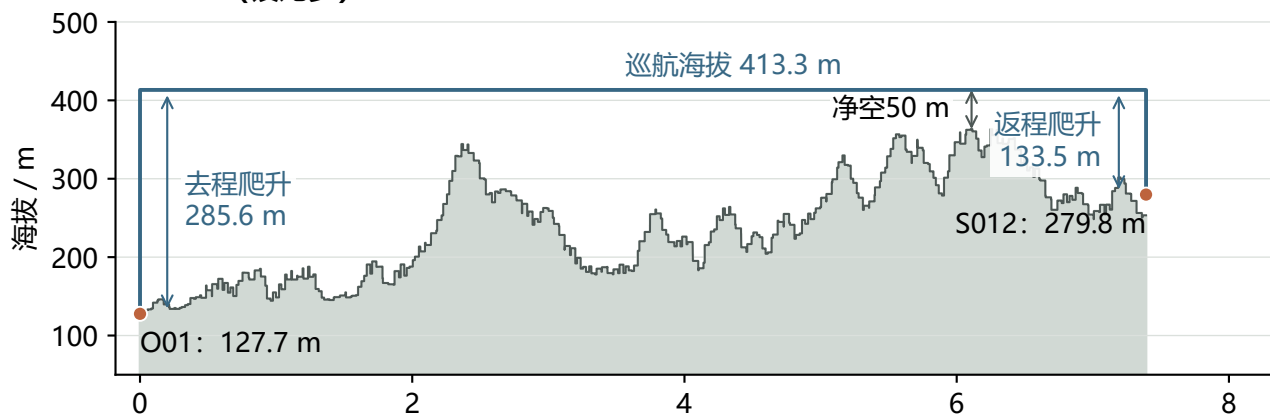
a S004 (大站)



b S008 (古麻、大站)



c S012 (镇龙乡)

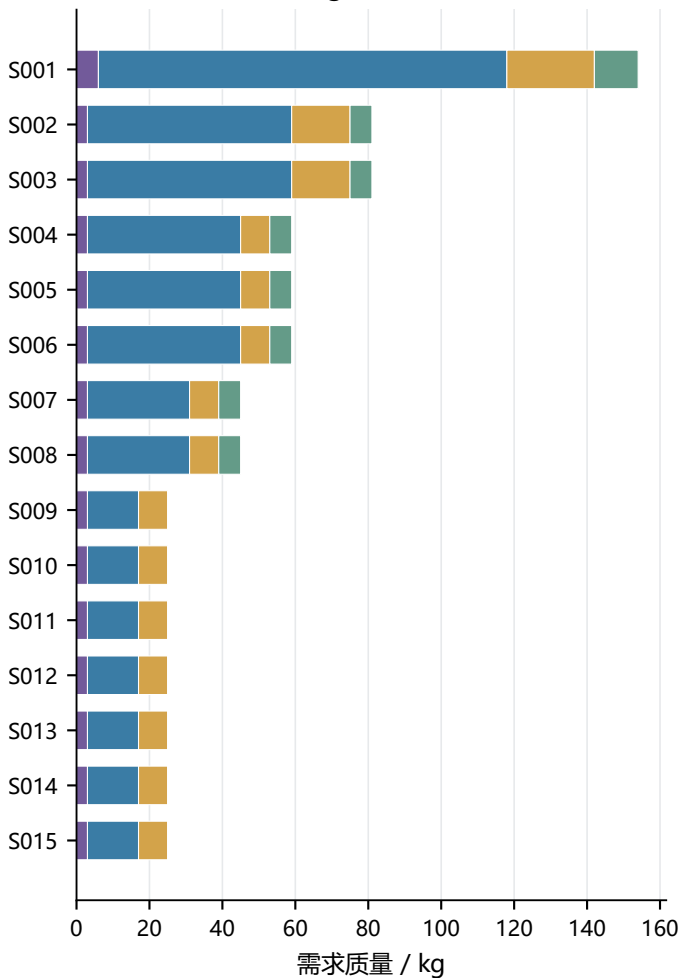


自调度中心起的水平距离 / km

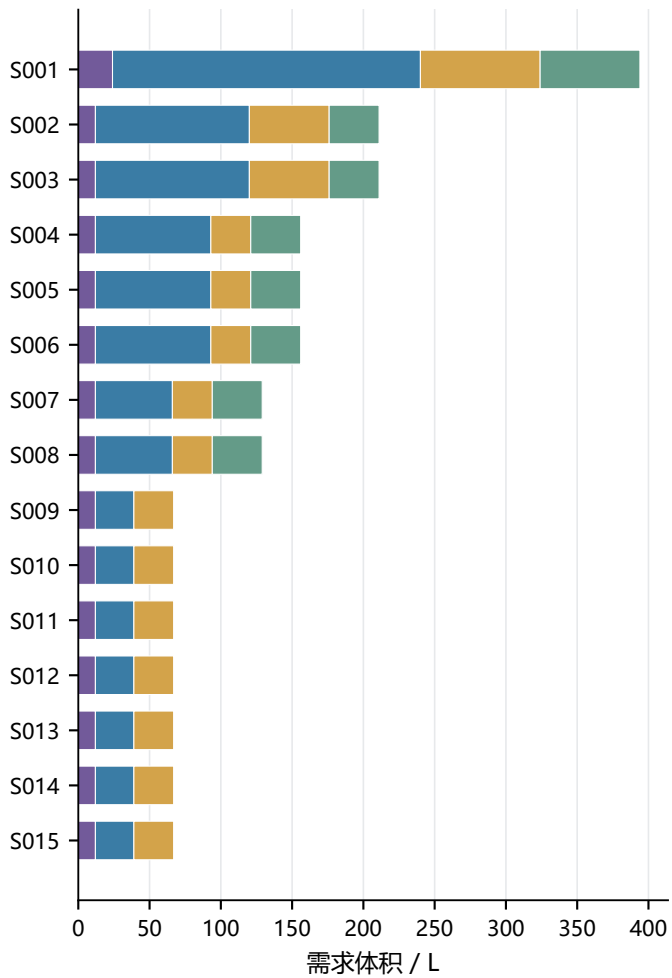
■ 相交像元地形上包络 — 飞行高度 ● 节点作业海拔

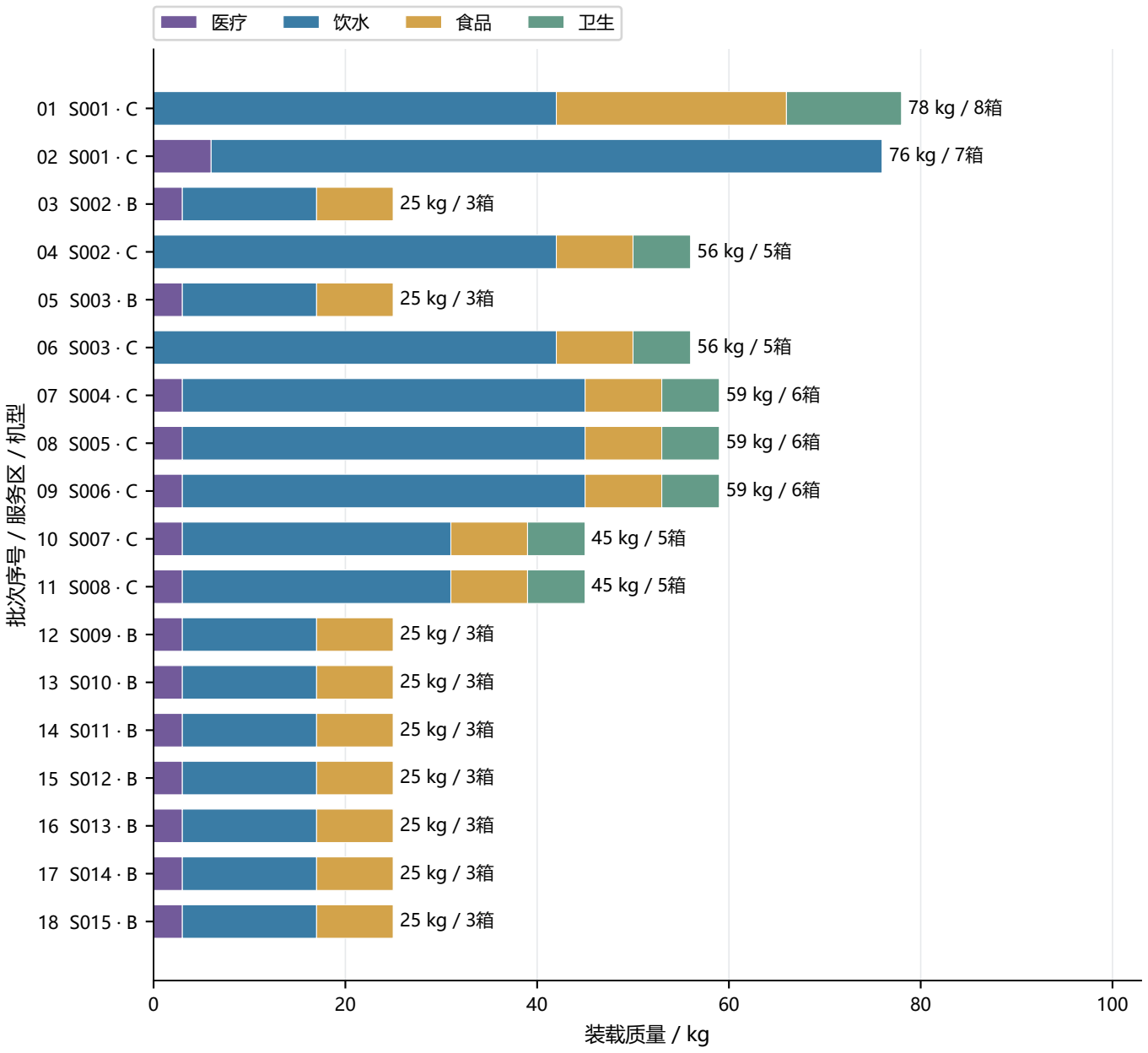


(a) 质量：总计 758 kg

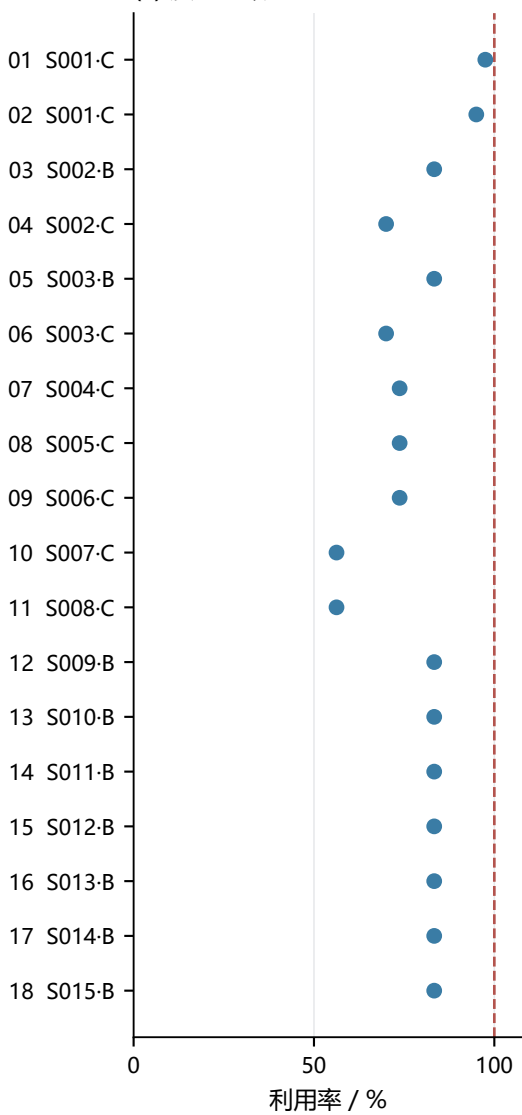


(b) 体积：总计 2011 L

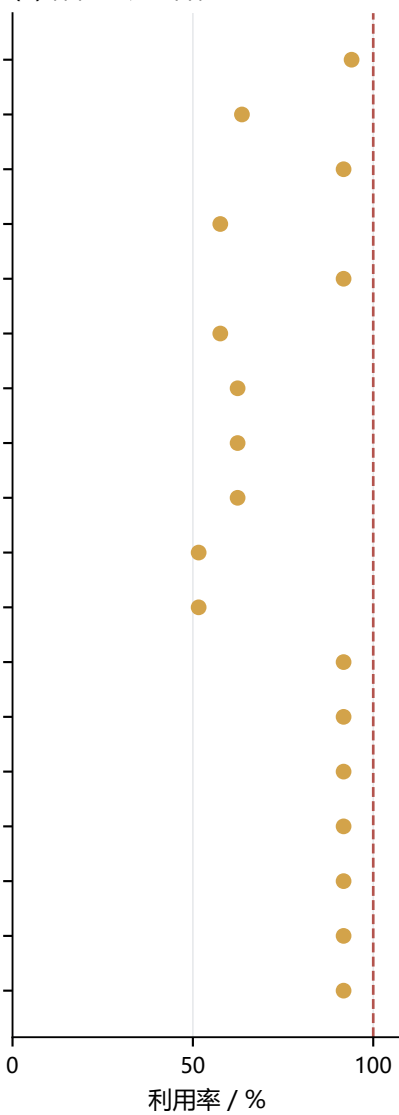




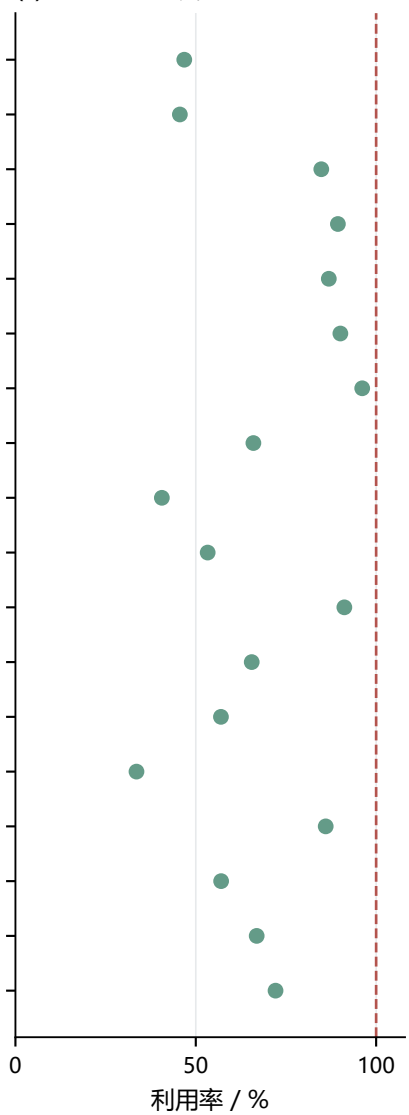
(a) 质量 / 额定载荷

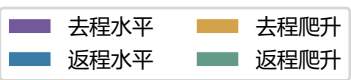


(b) 体积 / 装载容积

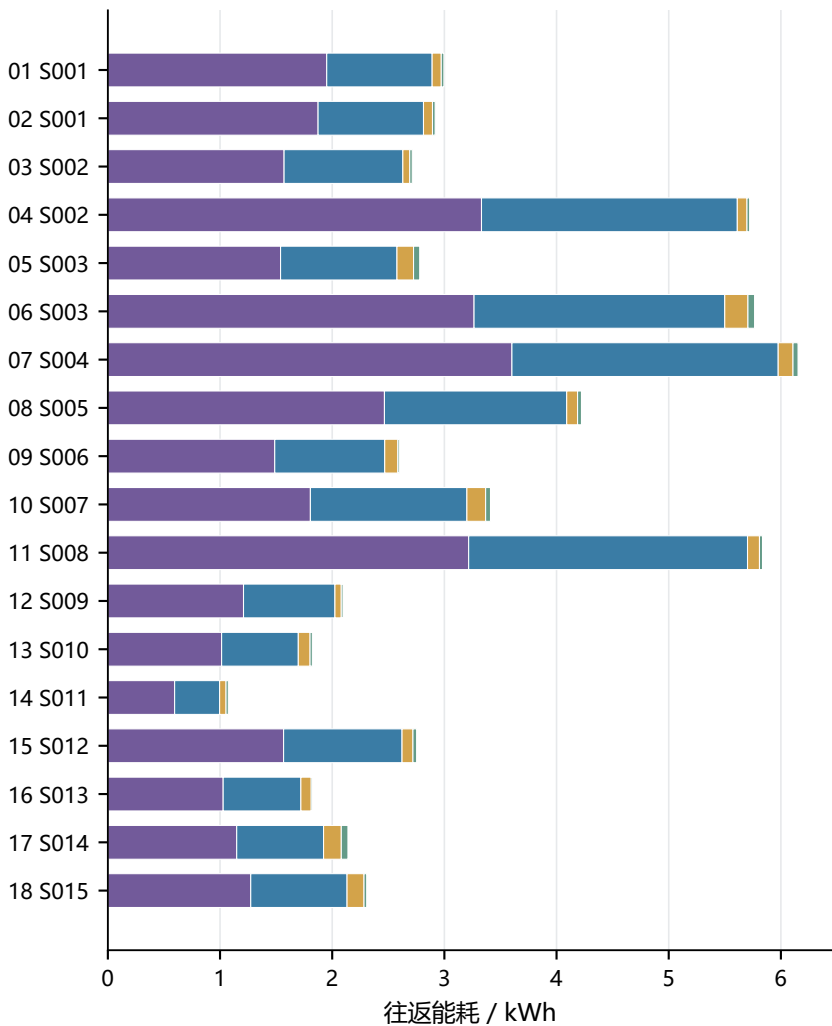


(c) 能耗 / 安全预算

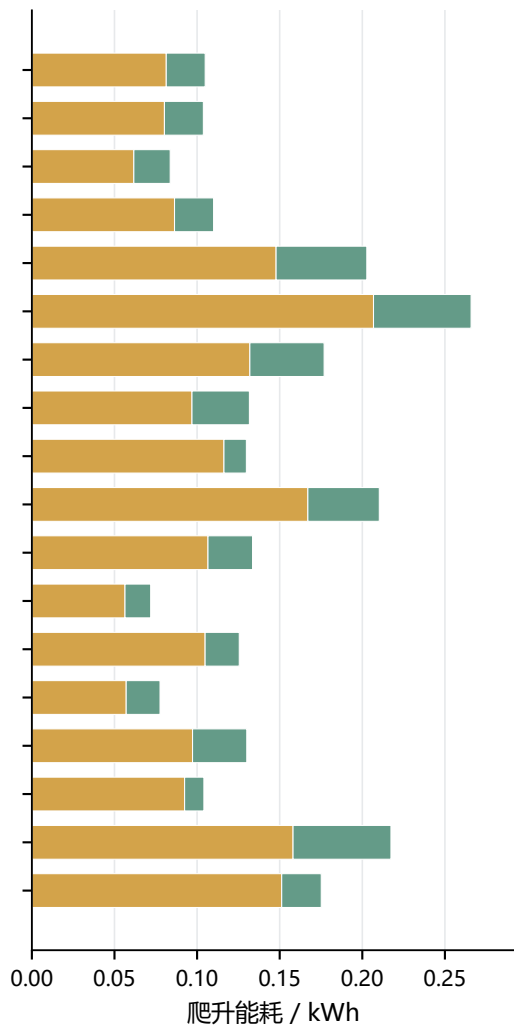


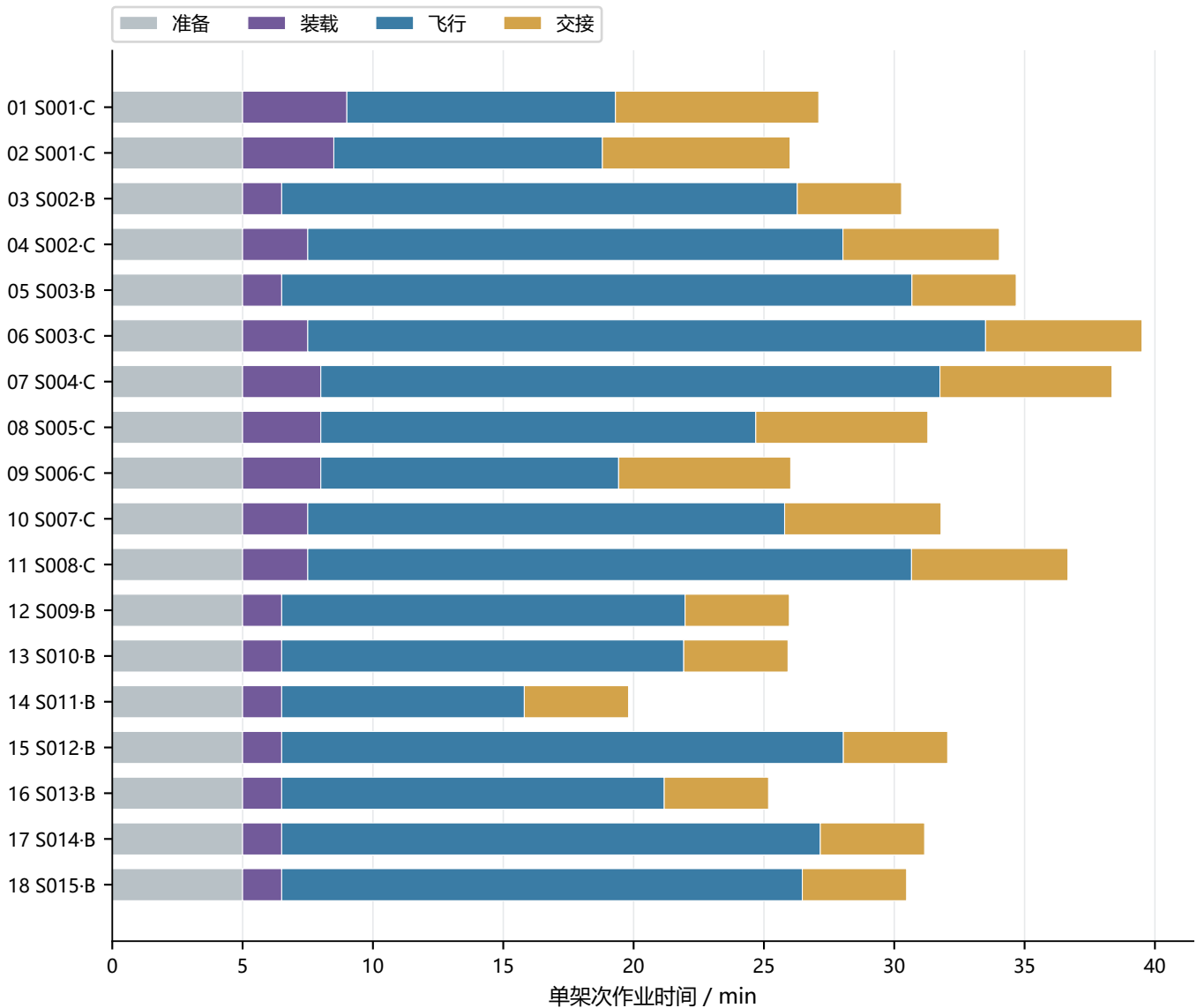


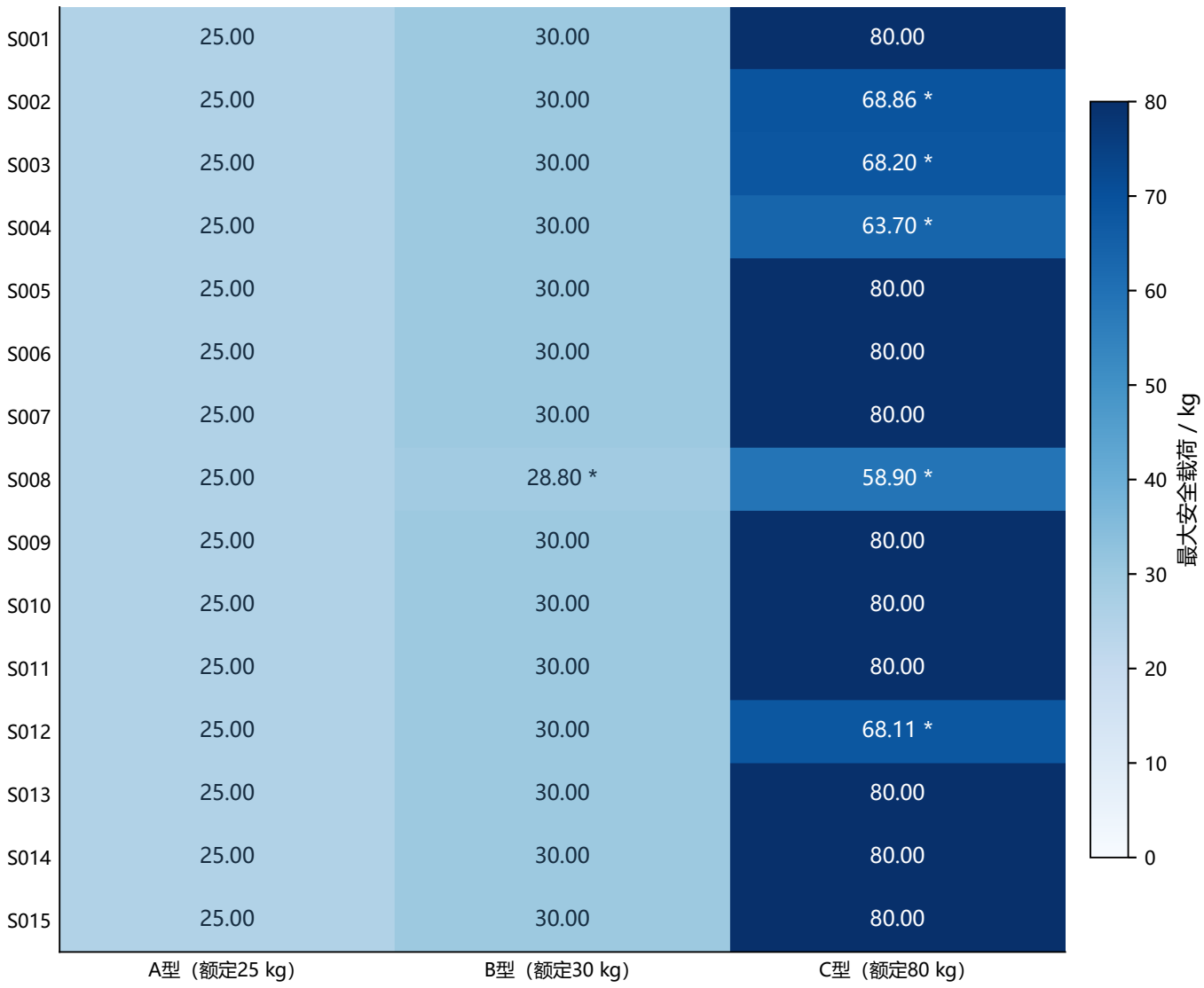
(a) 完整能耗分解



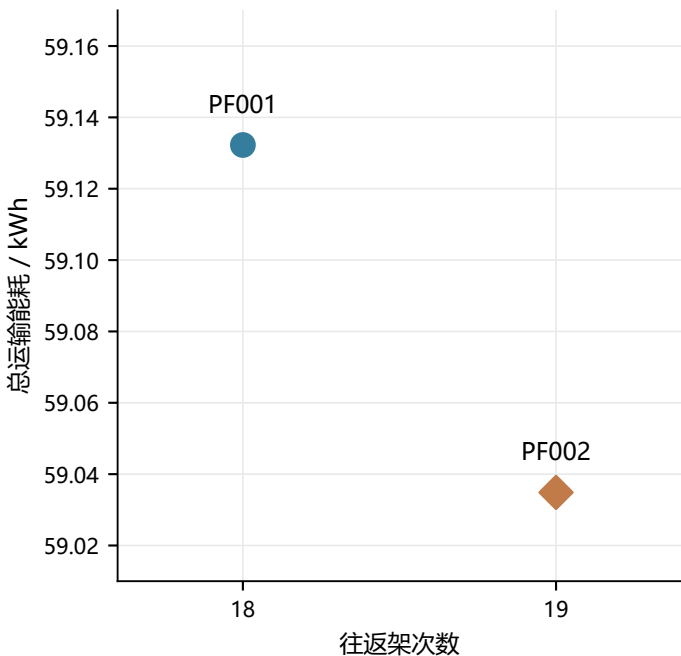
(b) 爬升分量放大



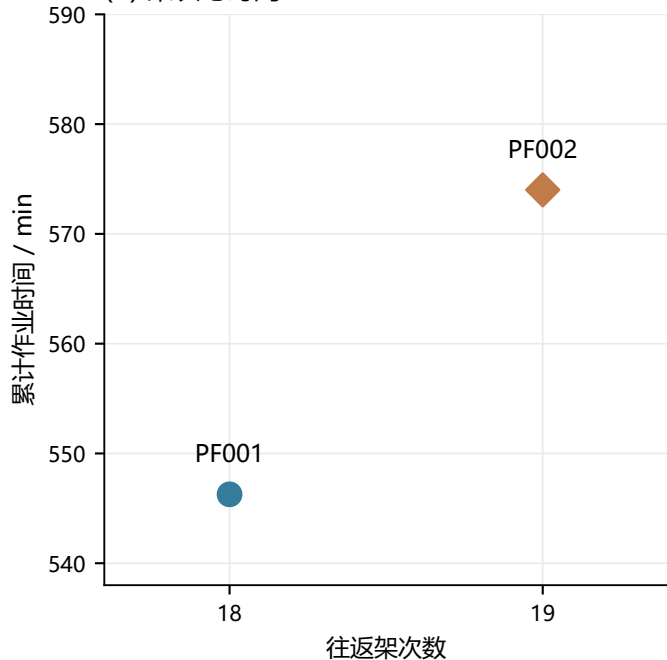


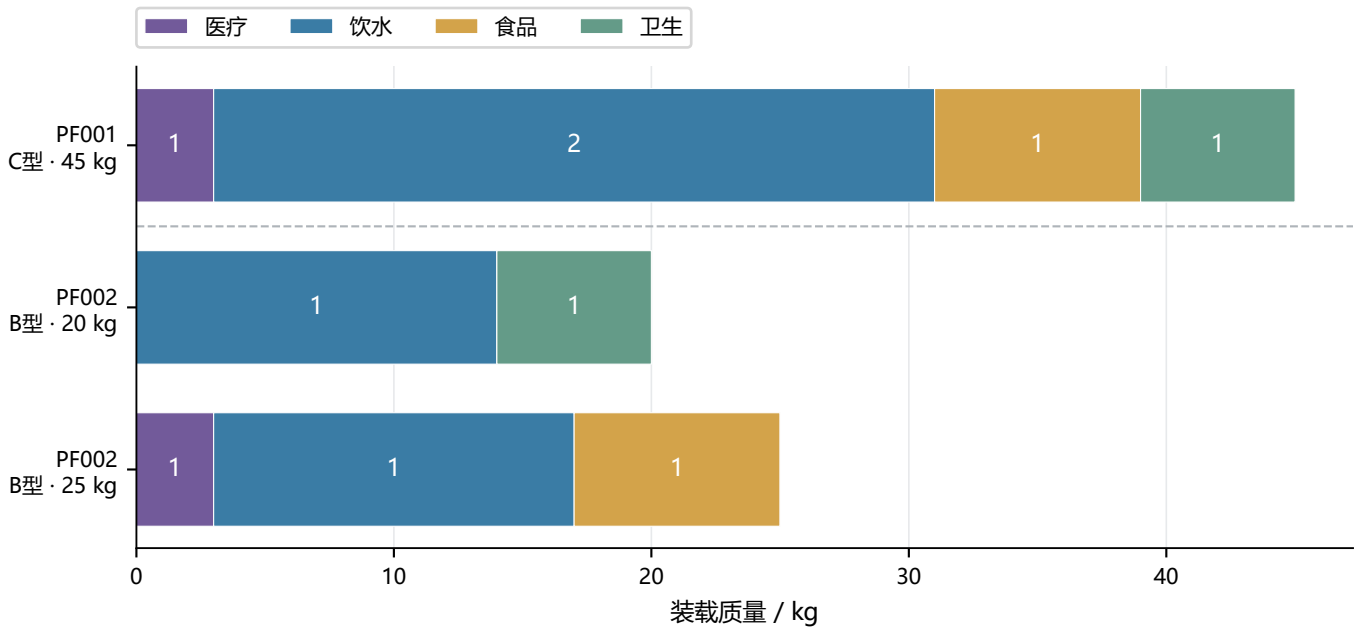


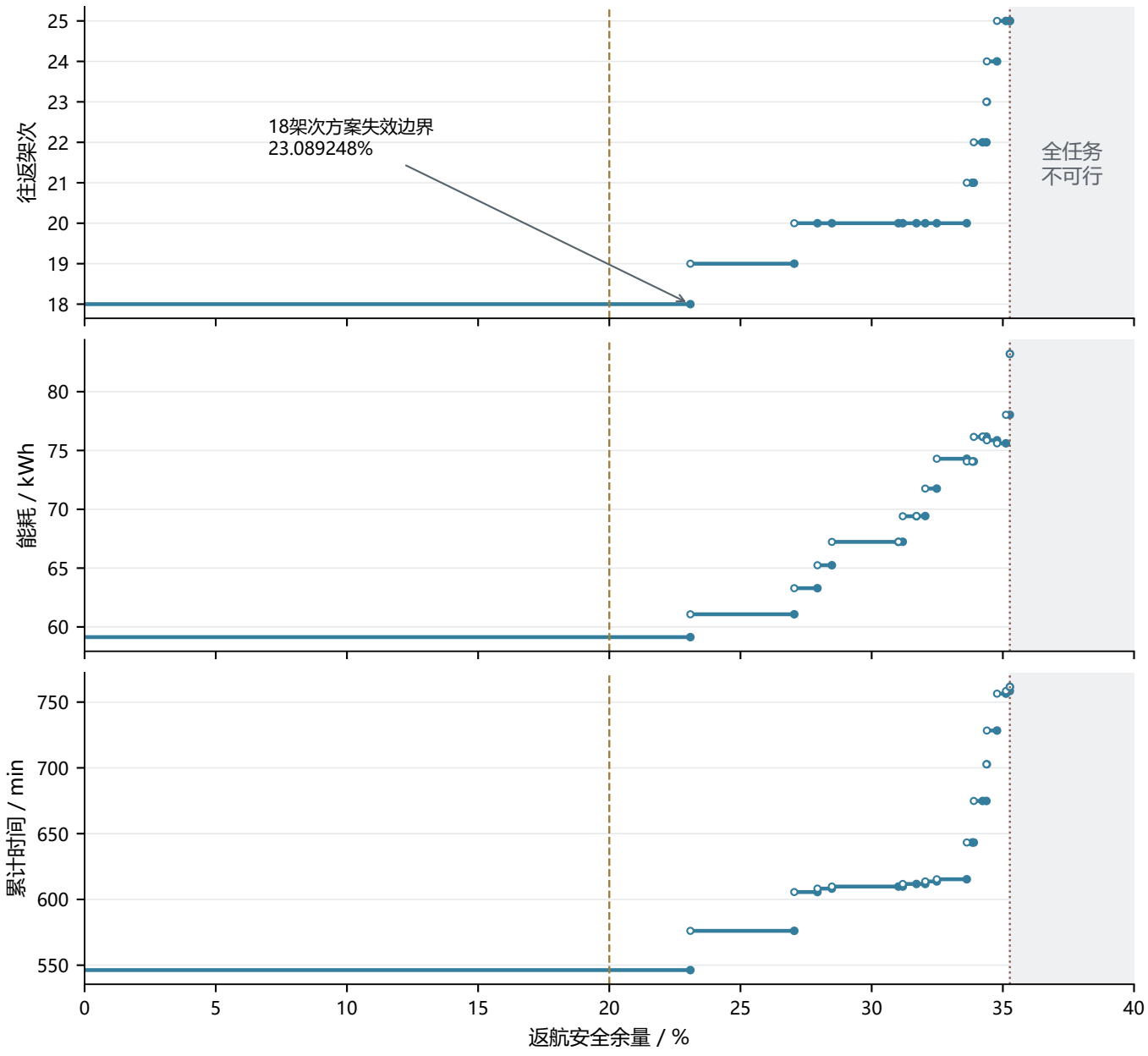
(a) 架次与能耗

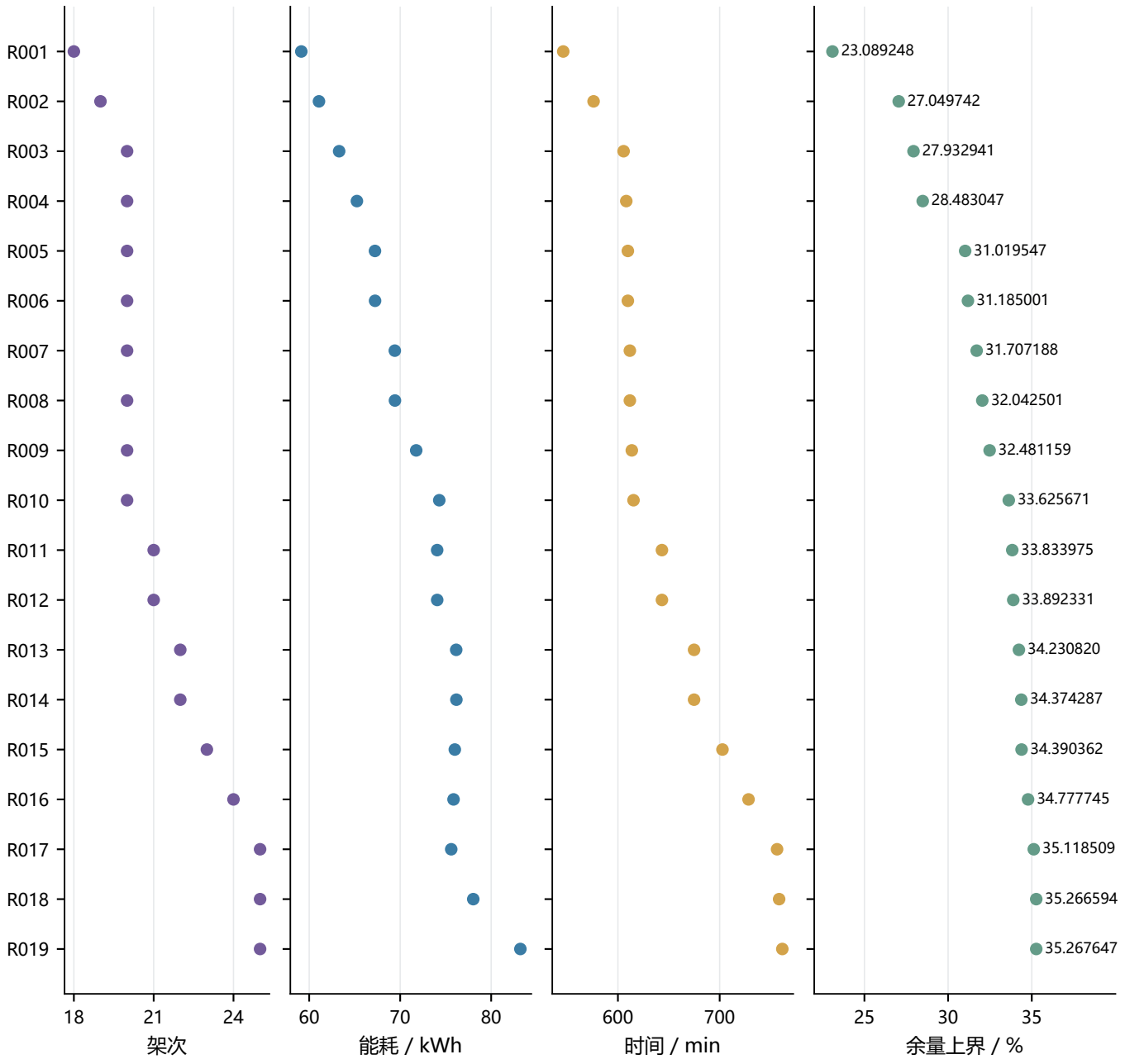


(b) 架次与时间

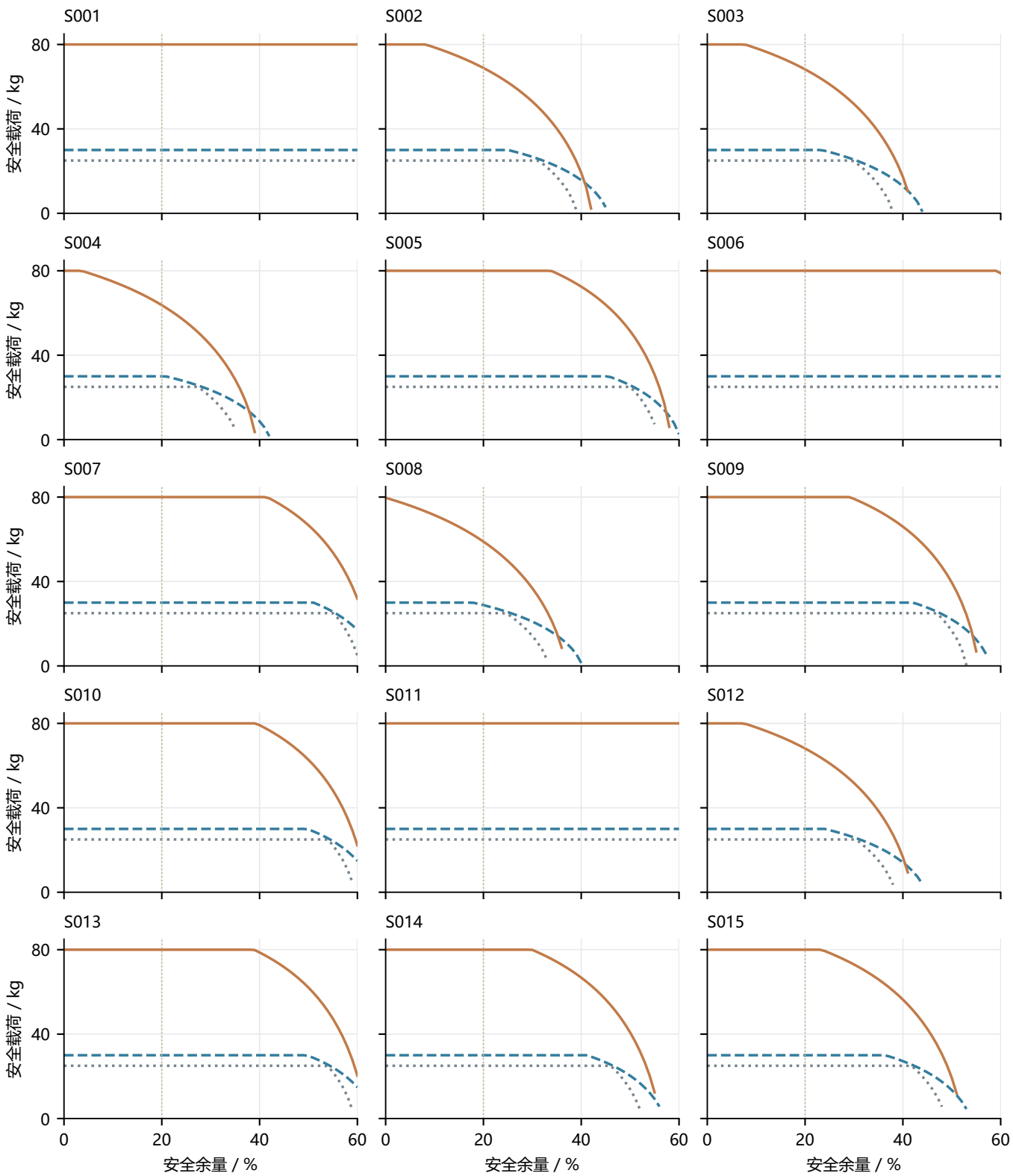




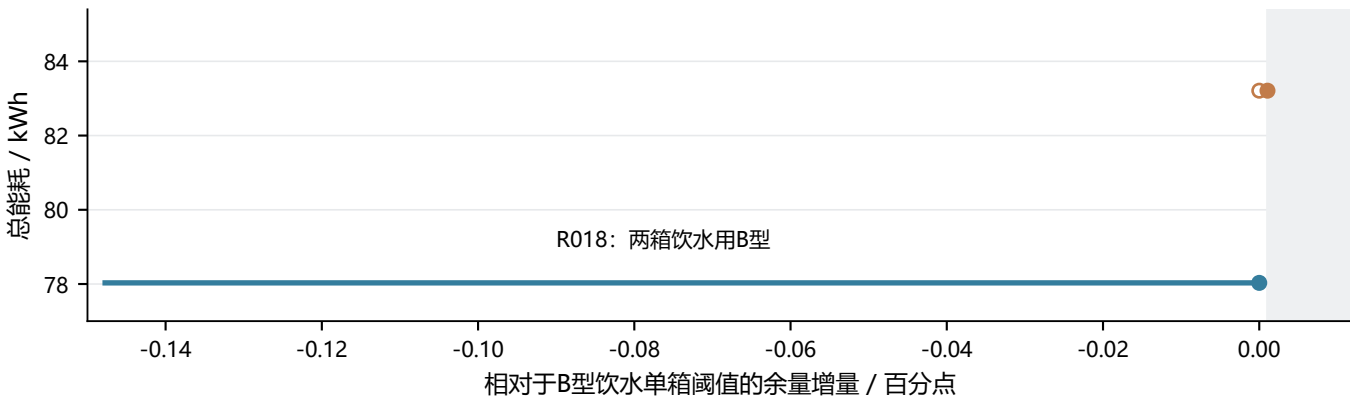




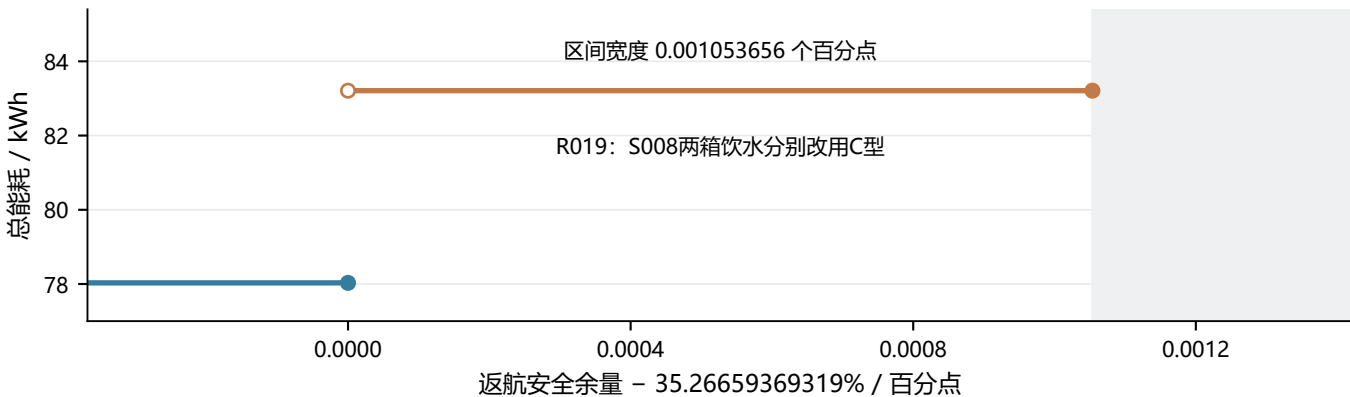
..... A型 - - - B型 — C型



(a) 最后两段的能耗变化

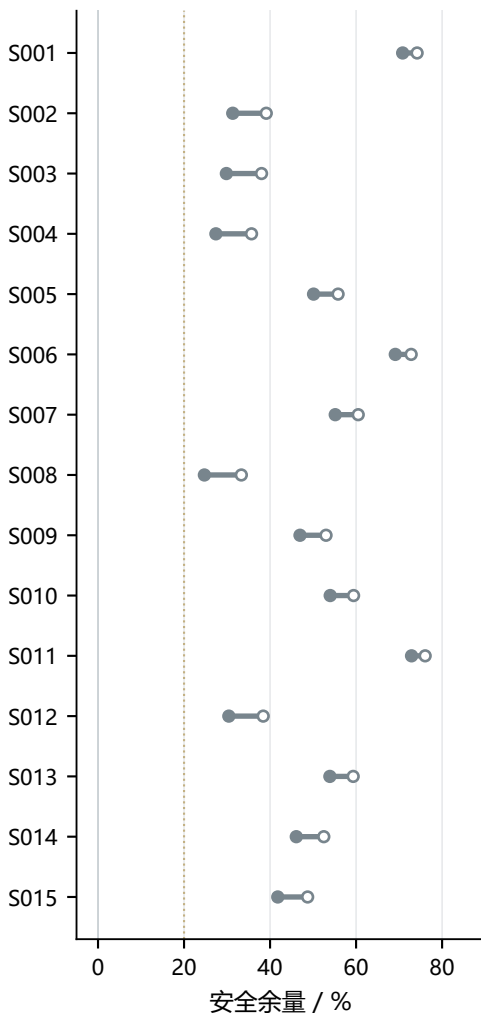


(b) R019窄区间放大：25架次保持不变

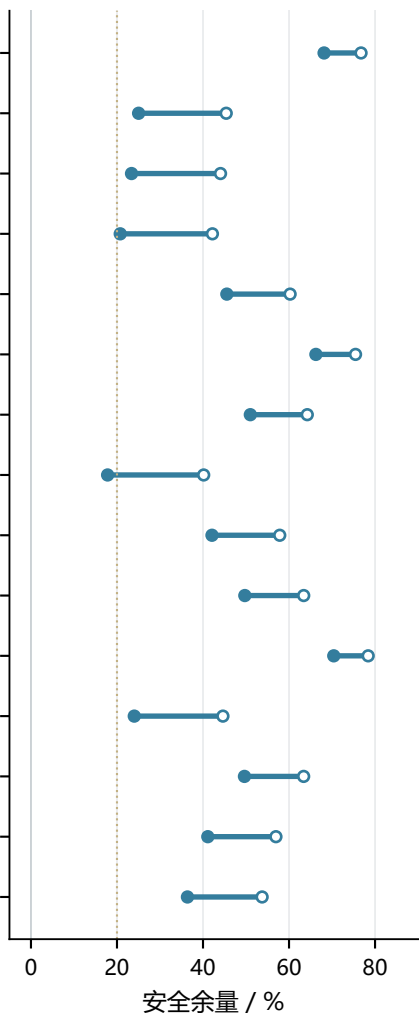


● 实心：额定满载上限 ○ 空心：空载往返上限

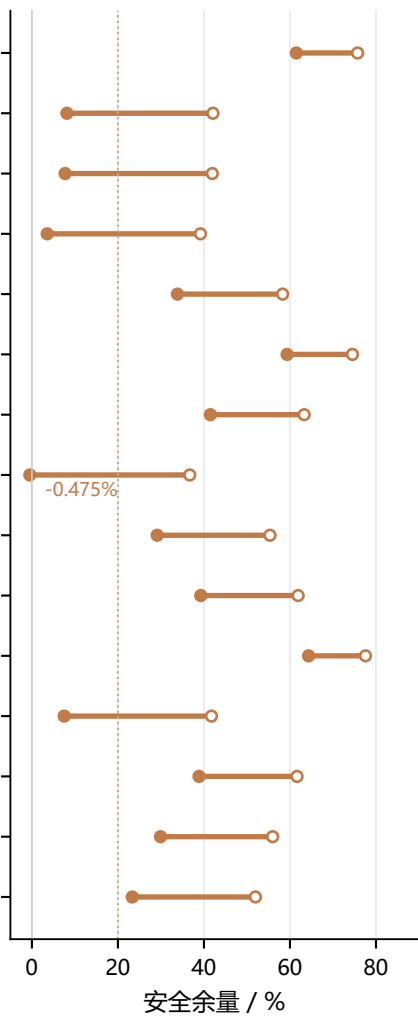
A型



B型



C型



题给输入与规则核验
16节点、3机型、80货箱、原始DEM

局部椭圆坐标 + 全相交DEM像元
距离、巡航高度、双向爬升、能耗与时间

枚举整箱数量组合
742个容量可行候选; 20%余量下732个候选

数量状态DP
按(N,E,T)字典序
基准18架次

非支配标签DP
逐区求解、全局合并
完整Pareto前沿2点

精确事件扫描
110个内部临界值
111区间 → 19方案段

恢复真实箱号 + 独立复核
逐箱唯一性、容量、能耗、端点、完整前沿与载荷
论文表格、图组与可回查源数据